

NETWORK

- A network is interconnection of two or more entities, or objects for sharing resources and information.
- Sharing(giving and getting) is fundamental aspect of networking.
- Networks are Everywhere.

Human Network

- Family Network:
 - most people belong to family network
 - people share their resources and information.
 - Sharing is bidirectional.
 - As Family grows, so does the network.
- Diagram

Peer Network

- Communities such as (Outside Family):
 - Restaurants
 - School
 - Store
 - Work

- **Restaurant Network**
- The Client and the Server
 - Customer -Client
 - Waiter works as a server
- **Contact Network-**
 - Finding the perfect job
 - Friends , Work associates, Teachers
 - Former Employee, A Friend's friend

COMPUTER NETWORK

A computer network is a **group of computer systems** and other computing **hardware devices** that are **linked together through communication channels to facilitate communication and resource-sharing** among a wide range of users.

OR

- Two or **more computers** that are **linked** in order to **share resources** (such as printers and CDs), exchange files, or **allow electronic communications**.
- The computers on a network may be **linked through cables, telephone lines, radio waves, satellites, or infrared light beams**.

Human Vs Computer N/W

Human Network	Computer Network
Human Network is a network of human beings.	Computer network is a interconnections of two or more computers
It consist of individuals ,organizations, schools, hospitals, work places, etc.	It consist of computers & connecting devices like as Hub or printer, etc.
Example: 1.Family network, Peer Network, Restaurant Network, Contact Network	Example : Internet

APPLICATIONS

- **1. Banking**
- **2. Video conferencing**
- **3. Marketing**
- **4. School**
- **5. Radio**
- **6. Television**
- **7. E-mail**
- **8. Companies**

Benifits of Computer Network

- **Sharing of Information**
- **Resource sharing**
 - **Peripherals:printer,scanner, speakers,etc.**
 - **Storage**
 - **Application sharing**
- **User Communication**
- **Facilitating Centralized Management**
 - **Managing S/w**
 - **Maintaing N/w**
 - **Backing up data**

Advantages/Benifits

- 1. File sharing: Computer network is that it allows file sharing and remote file access.** A person sitting at one workstation that is connected to a network can easily see files present on another workstation, provided he/she is authorized to do so.
- 2. Resource Sharing:** A computer network provides a **cheaper alternative by the provision of resource sharing.** All the computers can be interconnected using a network and just one **modem & printer can efficiently provide the services to all users.**

Advantages/Benifits

3. Inexpensive set-up:

Shared resources=> means reduction in **hardware costs**. **Shared files**=> reduction in **memory** requirement=>reduction in file **storage** expenses.

4. Flexible Handling: log on from anywhere and access his/her files. This offers flexibility to the user

5. Centralized Management- management of **resources** in the organization, **centrally** through **client server** architecture.

6. Backing up data: Creating **backup files** and **restoring** them becomes much **easier**.

7. E-mail Services

File sharing

- File sharing is the **primary feature** of network.
- Due to use of networks, the mark sharing of files becomes easier.
- File sharing requires a shared directory or disk drive to each) which many users can access over the network.
- When **many users are accessing the same file on the network**, more than one person can make changes to a file at the same time. They might both making conflicting changes simultaneously.

Printer sharing

- Printer sharing is beneficial to many users as they can share **costly & higher quality printers**.
- Printer sharing can be done in several different ways on network. The most common way is to use printer queues on server.
- The printer queue holds print job until any currently running print jobs are finished & then automatically send the waiting jobs to the printer i.e. printer connected to server.
- Another way to share printer on a network is that each workstations accesses the printer directly.

Application services

- You can also share application on a network.
- For example you can have a shared copy of **Microsoft office** or some other application & keep it on the network server.
- Another application services you can have on the network is shared installation i.e. contents of CD-ROM **copy to the server**, then **run the installation the installing application much faster & more convenient.**

E- mail services:

- E-mail is extremely valuable & important feature for communication within organization or outside the people in world.
- E-mail service can be used by user in two connecting or handling the e-mail interface between the two systems using gateway software that is part of the file-based e-mail system.
- A client-server e-mail system is one where an e- mail server containing the messages & handles all incoming & outgoing mail. It is more secure & powerful than file based e-mail system.

Remote access

- Using this feature user can access their file & e-mail, when they are travelling or working on remote location.
- It enables users to access to centralized application, stored private or shared files on LAN.

Internet & Intranet

- **Internet:** It is public network. This consists of thousands of individual networks & millions of computers located around the world.
- Internets have many different types of services available such as e-mail, the web & Usenet newsgroups.
- **Intranet:** It is private network or it is company's own network. Company use this feature for internal use.
 - For example: company establish its own web server, for placing documents such as employee handbooks, purchases form or other information that company publishes for internal use. It also has internet services such as FTP servers or Usenet servers.

Disadvantages:

- 1. Security concerns:** One of the major drawback of computer network is the security issues that are involved.
- 2. Virus and malware:** Viruses can spread on a network easily because of the interconnectivity of workstations.
- 3. Lack of robustness:** If the main file server of a computer network breaks down, the entire system becomes useless.
- 4. Needs an efficient handler:** The technical skills and knowledge required to operate and administer a computer network.